

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Pattern recognition [IT_4403]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze how discriminant functions influence the shape and position of decision surfaces for two-class normal distributions. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Explain the concept of Bayesian Decision Theory. How does it help in minimizing classification error? (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Compare and evaluate the performance of MLE, MAP, and Bayesian estimation methods in the context of limited data availability. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Explain the difference between parametric and non-parametric classification approaches. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Given two one-dimensional classes $\omega_1 = \{1, 2, 3\}$ and $\omega_2 = \{7, 8, 9\}$, use the Parzen Window method with a window width (h) = 2 to estimate the probability density function (pdf) for each class at the point $x = 4$. Based on the estimated densities, determine the most likely class for the sample point, and explain how changing the window size (h) influences the smoothness and accuracy of the pdf estimation. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Discuss the concept of the kernel trick in SVM. Explain how kernels such as Linear, Polynomial, and RBF enable SVM to handle non-linearly separable data. Give an example where applying a kernel improves classification accuracy. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Compare the Perceptron and Support Vector Machine (SVM) in terms of their optimization objectives, handling of non-linearly separable data, and the concept of margin maximization. (5)
- A) Provide suitable examples to support your comparison. Course Outcome: "CO4" / Cognitive Level: "Application"
- B) A training dataset has three class labels (Yes, No, Maybe) distributed as follows in a particular
- (5)

node:

Yes = 4, No = 3, Maybe = 3

Tasks:

1. Calculate the Entropy for the node.
2. Calculate the Gini Index for the same node.
3. Interpret the results - explain which measure Entropy or Gini indicates a purer node and why.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Perform Complete Linkage Hierarchical Clustering using the following Euclidean distance matrix. (5)
Identify which clusters merge at each iteration and construct a dendrogram based on the complete linkage distances.
- A)

	1	2	3	4	5
1	-	4.0	11.7	20.0	21.5
2	4.0	-	8.1	16.0	17.9
3	11.7	8.1	-	9.8	9.8
4	20.0	16.0	9.8	-	8.0
5	21.5	17.9	9.8	8.0	-

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Compare the DBSCAN and K-Means clustering algorithms in terms of their use of centroids, (5)
handling of arbitrary-shaped clusters, and sensitivity to noise and outliers. Discuss with suitable examples.

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4403:Pattern recognition

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4403:Pattern recognition

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	50(100%)

4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)